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DataBase Management System
Assignment - 02

B.Tech CSE Core (A)

Submitted to :
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Fill in the Blanks:-

1. The Entity Relationship model visually represents how entities relate in database, offering a clear blueprint of the data's organization structure.
2. In the Relational model, data is organized into tables, with relationship between entities defined through foreign key constraints.
3. The Object-oriented model treats data as a collection of objects, each with unique attributes & methods, mirroring real-world entities & interactions.
4. Integrity constraints ensure the reliability of data by offering rules that must be adhered to, maintaining the accuracy information.

Section-B

In a healthcare system, discuss how data abstraction can simply the interaction b/w heathcare professionals & patient records.

Data abstraction means hiding complex dB details & showing only necessary information to users.

In health care system, this helps doctors, nurses & hospital staff access patient information easily without dealing with complex structure of the dB.

For e.g a doctor only needs to see patient details such as name, age, medical history, test reports & prescribed medicines. Doctor doesn't need to know how the data is stored in dB tables. Data abstraction hides these technical details & presents the information in a simple form.

Similarly, a receptionist may only need basic information such patient name, appointment time & appoint doctors.
They do not need access to medical records or treatment history.
Data abstraction allows diff. users to see only the information relevant to their role.

→ Therefore Data abstraction makes healthcare system easier to use, improves efficiency & helps healthcare professionals quickely access important patient information.

3. Explore the potential benefits & challenges of implementing data abstraction in healthcare system.

→ Benefits:-

→ Improved Data Security :- Sensitive patient information is protected. Only authorized users can access medical records.

→ Ease of Use :- Staff can easily access the information without understanding dB.

→ Faster Decision Making :- Doctors can easily view patient history & provide treatment.

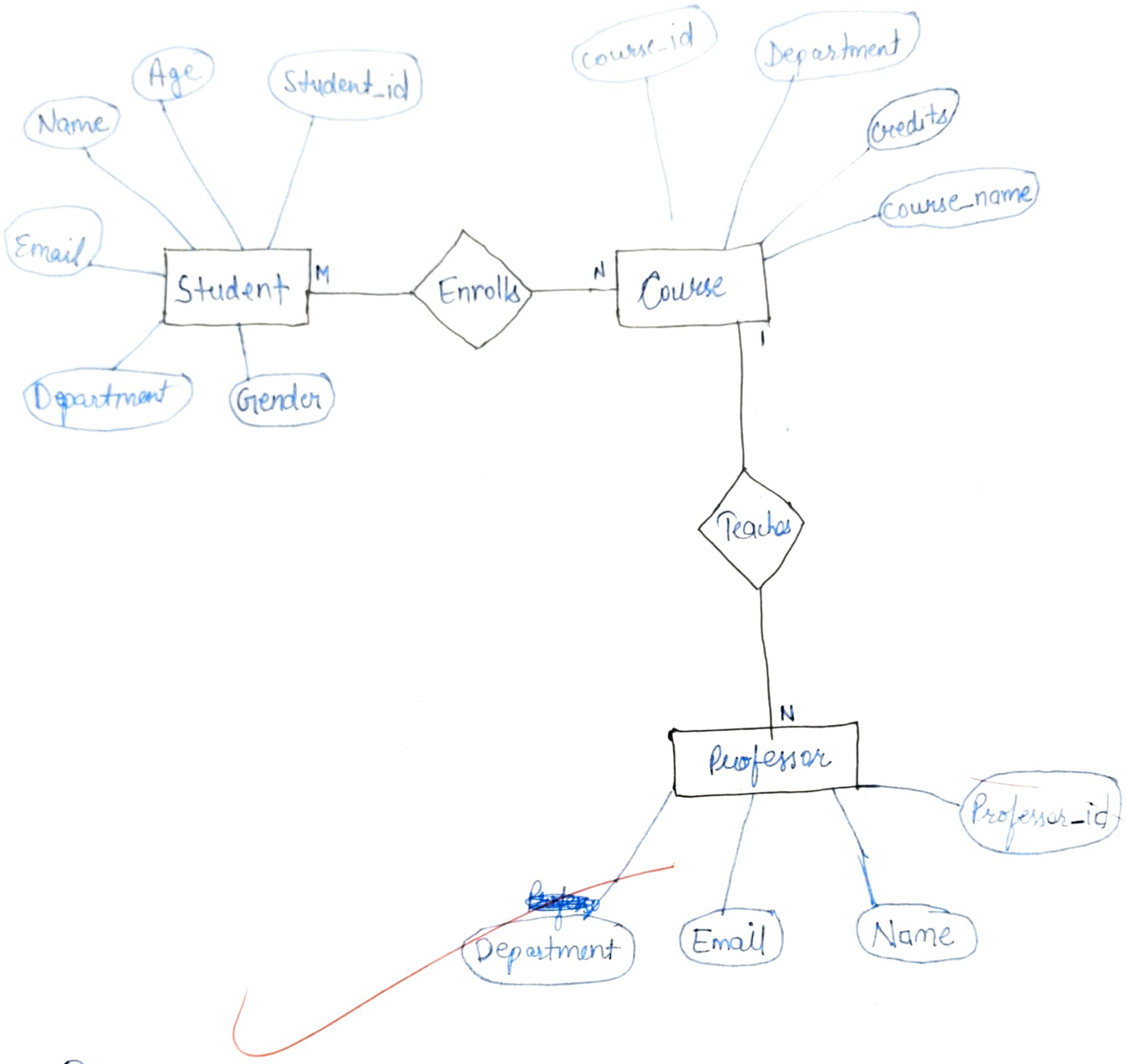
Better Data organization :- Records are stored in systematic & structure format.

Flexibility in System Design :- If DB structure changes, users are not affected bcz they only see the abstract view.

Challenges :-

1. Complex System design
2. Development time
3. Security Management
4. Maintenance Effort.

4) Design a DB for university using E.R model.



Entities :- Student, Course, Professor

Attributes :- Student_id
Student_name
Age
Email
department
Gender

} Student

2. Course_id
Name
Credits
Department } Course

3. Professor_id
Name
Department
Email } Professor

→ Student_id, course_id, Professor_id are primary key.

Relationship b/w Entities:-

1. Student — Course Relationship (Enrollment)

→ A student can ~~enroll~~ enroll in multiple courses
→ A course can have multiple students.
→ This is a Many to Many Relationship.

2. Professor — Course Relationship (Teaching)

→ A professor can teach multiple courses.
→ Each course is taught by one professor.
→ This is One to Many relationship